Date

Expt. No. 7

Page No. 21

Java Servlet

Aim:- To study server-Jaiside programming using Java servlet & create a simple web applicatn that handles an HTTP request & generates a response.

Theory:- A servlet is a Java class that runs on a server to processes client requests to generate dynamic web responses. Apache Tomcat acts as the servlet container & manages the servlet lifecycle.

The HttpServlet class is commonly extended to create a servlet. The doGet() method handles GET requests, while doPost() handles POST requests. The HttpServletRequest object receives client data, & HttpServletResponse generates the response.

Code File: HelloServlet.java.

```
import java.io.IOException;
import java.io.PrintWriter;
import jakarta.servlet.ServletException;
import jakarta.servlet
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;

@WebServlet("/Hello")
public class HelloServlet extends HttpServlet {
```

Teacher's Signature

Output

http://localhost:8000/servlet/hello

Hello Students!

Welcome to servlet programming.

Expt. No. _____ Date _____

Page No. 22.

```
@Override
protected void doGet(HttpServletRequest request,
        HttpServletResponse response)
throws ServletException, IOException {
response.setContentType("text/html");
PrintWriter out = response.getWriter();

out.println("<h1>Hello Students!</h1>");
out.println("<h2>Welcome to servlet programming
                                    </h2>");
}
}
```

Procedure

1. Create a Java web application in Tomcat
2. Add HelloServlet.java to the project.
3. Deploy the application on Apache tomcat
4. Open the URL in a web browser

Viva Question

Q1 What is a Servlet?

Ans A servlet is a java class that processes client requests & generates dynamic web ~~proces~~ responses.

Q2 What is Apache Tomcat?

Ans Apache Tomcat is a web server & servlet Container used to run java web applications

Q3 What is the doGet() method?

Ans To doGet() method handles HTTP GET request from clients

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Q4 What is the use of @webservlet?

Ans The @webservlet annotation maps a servlet to a particular URL pattern.

Q5 What are HttpServletRequest & HttpServletResponse?

Ans HttpServletRequest represents the incoming request, while HttpServletResponse is used to generate the server's response.

Date

Expt. No. 8.

Page No. 24.

Java HTML to JSP

Aim:- To understand JavaServer Pages (JSP) & convert an HTML page into a JSP pages to display student information dynamically.

Theory :- JSP (JavaServer pages) is a server-side technology used to create dynamic web pages. It allows HTML to be combined with JSP elements & Expression Language (EL) to display data generated on the server.
A servlet processes the request & handles application logic, while JSP displays the result. The JSP container translates a JSP page into a servlet, compiles it, & executes it to generate the response.

Code:- (Student.jsp)

```
<%@ page contentType = "text/html; charset
        = UTF-8" %>
<!DOCTYPE html>
  <html>
    <head>
      <title> Student Information </title>
    </head>
  <body>
    <h1> Student Information </h1>
    <form action = "hello" method = "post">
    name: <input type = "text" name = "name">
                                <br><br>
Email: <input type = "email" name = "Email"> <br><br>
Course: <input type = "text" name = "Course"> <br><br>
Age: <input type = "number" name = "age"> <br><br>
```

Teacher's Signature

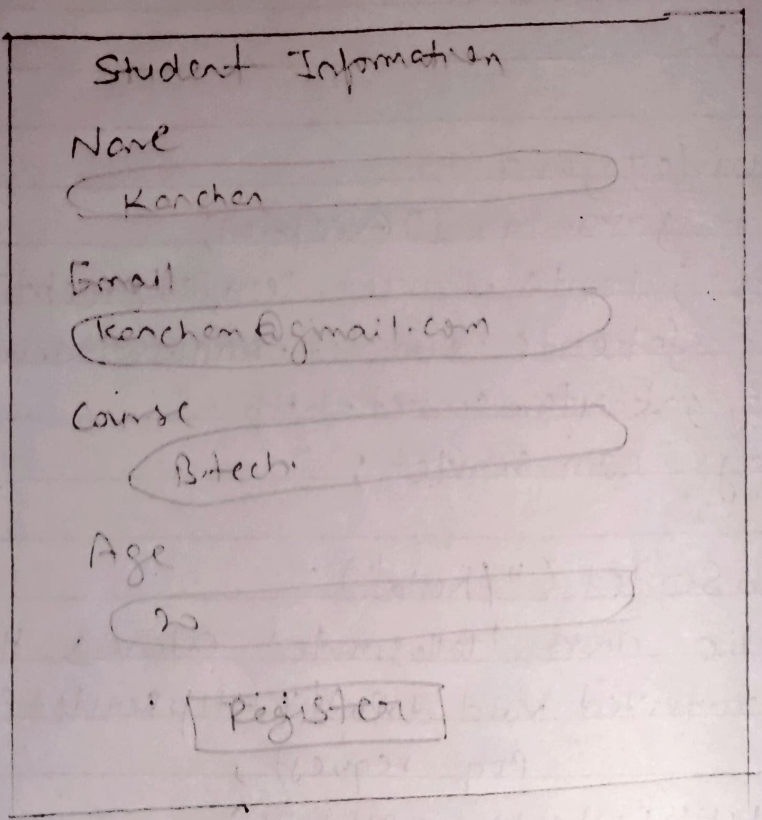
```
    <input type="submit" value="Register">
  </form>
</body>
</html>
```

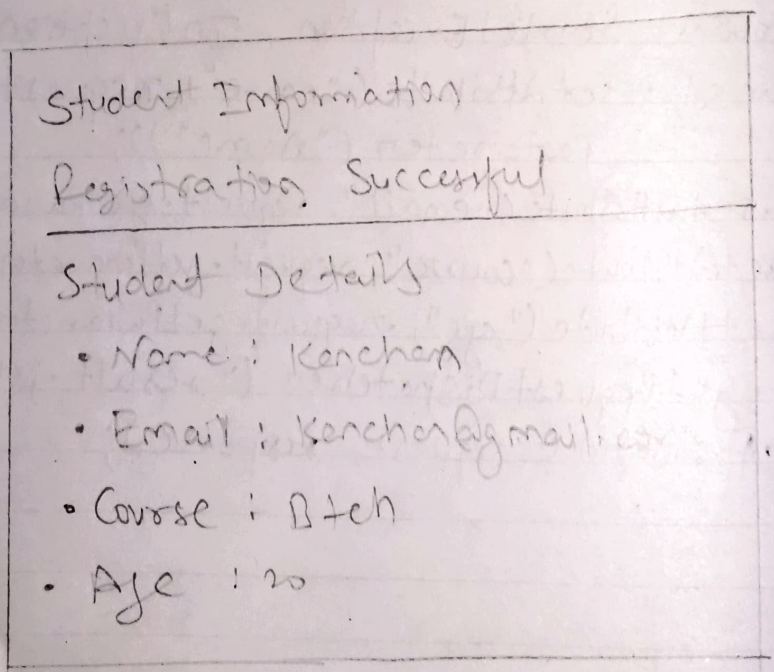
helloservlet.java

```
import java.io.IOException;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.*;
package com.servlet;

@WebServlet("/Hello")
public class HelloServlet extends HttpServlet {
    protected void doPost(HttpServletRequest
            Req request,
        HttpResponse response)
    throws ServletException, IOException {
request.setAttribute("name", request.get
        Parameter("name"));
request.setAttribute("email", request.getParameter("email"));
request.setAttribute("course", request.getParameter("Course"));
request.setAttribute("age", request.getParameter("Age"));
request.getRequestDispatcher("result.jsp")
    .forward(request, response);
  }
}
```

Output


Student Information
Name
Kanchan
Email
Kanchan@gmail.com
Course
Btech
Age
20
Register


Student Information
Registration Successful
Student Details
• Name : Kanchan
• Email : Kanchan@gmail.com
• Course : Btech
• Age : 20

Date

Expt. No. Page No. 26.

result.jsp

```
<%@ page contentType = "text/html; charset
        = UTF-8" %>
<!DOCTYPE html>
<html>
<head>
  <title> Registration Result </title>
</head>
<body>
    <h1> Registration Successfull </h1>
  <hr>
  <h3> Student Details </h3>

    <p> Name : ${name} </p>
    <p> Email : ${email} </p>
    <p> Course : ${course} </p>
    <p> Age : ${age} </p>
</body>
</html>
```

Procedure

1. Place student.jsp & result.jsp inside the project's src/main/webapp directory.
2. Place HelloServlet.java inside src/main/java/com/
3. Build & Deploy the project on Apache tomcat server.
4. Open http://localhost:8080/servlet/student.jsp.
5. Enter student details & click register. The servlet processes the data & forwards the request to result.jsp.

Teacher's Signature

Date

Expt. No. Page No. 27.

Viva Question

Q1. What is JSP ?

Ans JSP stands for JavaServer pages. It is used to create dynamic web pages on the server.

Q2. What is the difference between servlet & JSP?

Ans A servlet mainly handle requests processing, while JSP mainly handles presentation & displays data.

Q3. What is Expression Language (EL)?

Ans EL provides a simple way to access data in JSP, such as ${name}.

Q4. What is the use of request.setAttribute()?

Ans It stores data in the request scope so that another resource, such as a JSP, can access it.

Q5. What does forward() do?

Ans It forward the same request & response to another server-side resource, such as result.jsp.

Teacher's Signature